Re-processing Perturb-Seq datasets from FASTQ

First Results for the Unified Pipeline

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All Hands Meeting

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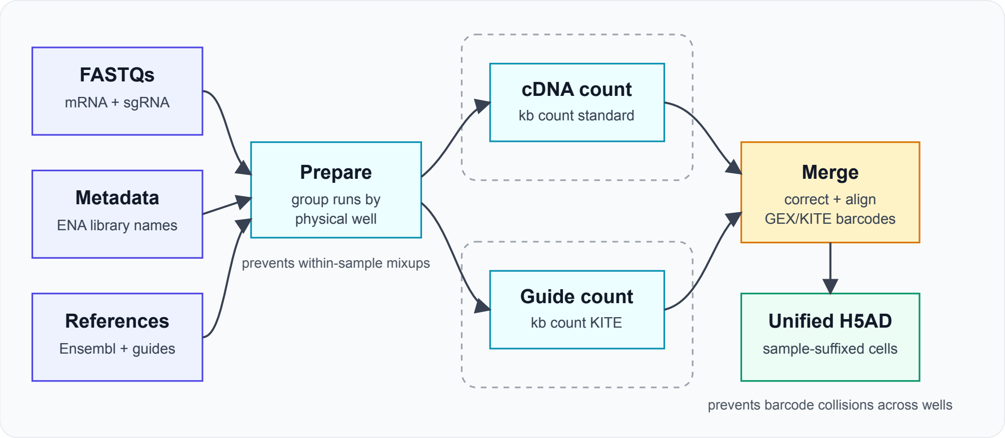
Why re-process from FASTQ?

- **Consistency:** scPerturb datasets are usually deposited as processed H5AD files, but they come from different pipelines, references, and filtering choices.
- **Comparable outputs:** Re-processing gives us a common count matrix layout, common metadata expectations, and a consistent way to attach guide counts.
- **Quality control:** Raw reads let us apply the same checks across studies instead of inheriting every author-specific decision.
- **Catalogue expectation:** Users expect harmonised data in the Perturbation Catalogue, not only links to heterogeneous processed files.

Why Kallisto + Bustools?

- **Speed:** Kallisto + Bustools is much faster than Cell Ranger while giving very comparable expression quantification.
- **Resource fit:** We have adequate, but not unlimited, compute and storage. The pipeline must work within those limits.
- **Full catalogue scale:** We want to process all datasets we have, and be able to reprocess them if the pipeline improves.
- **Future-proofing:** New perturb-seq datasets are expected to be very large, so the pipeline needs to scale well rather than depend on throwing more hardware at each run.

Pipeline Architecture



Development Issues Solved

- **ENA FASTQs missing technical reads:** Historical ENA FASTQs for this dataset do not include the barcode/UMI reads. Current workaround: SRA download with technical reads included. We notified ENA; they are aware and plan to fix it, so future runs may use ENA directly.
- **Sample grouping:** SRRs and lanes must be grouped by physical well before counting. This is currently parsed from ENA library names; later versions may autodetect this.
- **Barcode collisions:** Final concatenation adds sample suffixes so identical 10x barcodes from different wells remain distinct.
- **GEX/KITE matching:** Guide barcodes are not exactly the same as expression barcodes in this dataset. The pipeline applies the required KITE barcode correction before alignment.

Pipeline Execution

```
nextflow run main.nf \
  -profile slurm,singularity \
  --fastq_dir $HPS_PATH/perturb_seq_fastq/SAMN40972597 \
  --metadata_tsv $HPS_PATH/perturb_seq_fastq/SAMN40972597/ena_metadata.tsv \
  --outdir $HPS_PATH/perturb_seq_fastq/results \
  --chemistry 10xv3 \
  --transcriptome_fa $HPS_PATH/cache/reference/Homo_sapiens.GRCh38.dna.primary_assembly.fa.gz \
  --gtf $HPS_PATH/cache/reference/Homo_sapiens.GRCh38.111.gtf.gz \
  --features_tsv $HPS_PATH/perturb_seq_fastq/SAMN40972597/features.tsv
```

- The sample sheet is generated automatically from ENA metadata.
- mRNA and sgRNA workflows run in parallel per sample.
- Guide counts are merged from unfiltered KITE output, then aligned to filtered mRNA cells.

First Dataset: Nadig 2025 Jurkat

- The unified FASTQ-to-H5AD pipeline now runs end-to-end on the EBI Slurm cluster.
- The first full dataset has been processed: Nadig 2025 Jurkat.
- The comparison was run against the author-provided H5AD.
- Analysis compares shared cells and shared genes after barcode and gene-name alignment.
- The main question: does the reprocessed dataset preserve expression signal, and how many guide calls match the original labels?

Short answer: expression alignment is excellent; guide recovery is currently limited by strict dual-probe calling.

First Dataset: Scale and Runtime

Input scale

- Full raw-data project: 2.618 TB and 2,688 reported FASTQ files.
- Jurkat subset processed here: 1,792 SRR runs, reported as 1,792 FASTQ files.
- Split by modality: 896 mRNA runs and 896 sgRNA runs.

Pipeline run

- Wall time: **1h 18m**.
- Total compute: **504 CPU-hours**.
- Near-terabyte input processed with modest total compute.
- This is efficient enough to make catalogue-scale reprocessing realistic.

Expression Agreement: Summary

Metric	Pearson r
Total UMI counts per cell	0.9997
Detected genes per cell	0.9999
Mean expression per gene	0.9588
Gene dropout rate	0.9598

- Cell-level expression metrics are almost identical to the author-processed dataset.
- Gene-level expression and dropout also show strong agreement.
- The main total-count difference is scale, not loss of biological structure.

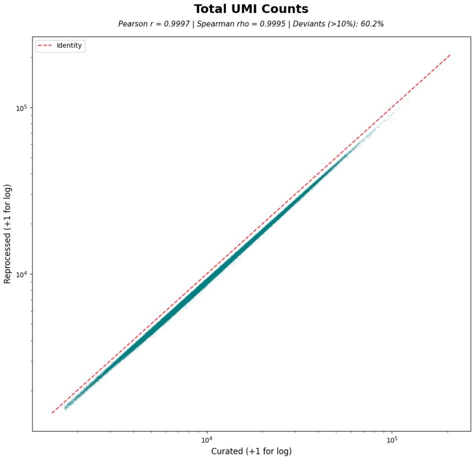
Expression Vectors Are Preserved

- For each shared cell, the comparison takes the curated and reprocessed expression vectors over the same shared genes.
- It then computes a Pearson correlation between those two vectors for that cell.
- This is done on a random sample of up to 10,000 shared cells after the same preprocessing.

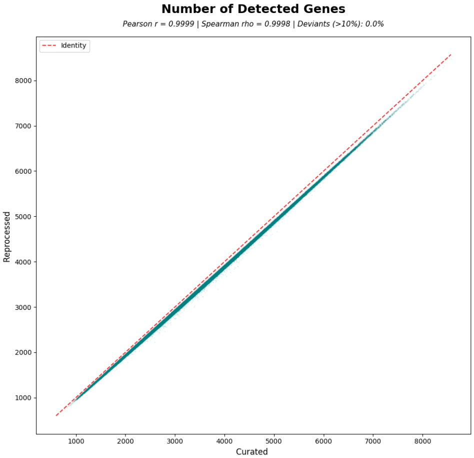
Cell-wise expression vector correlation	**Pearson r**
Median across sampled cells	0.9597
Mean across sampled cells	0.9594

Interpretation: the same cells have very similar expression profiles after reprocessing.

Cell-Level Expression Agreement

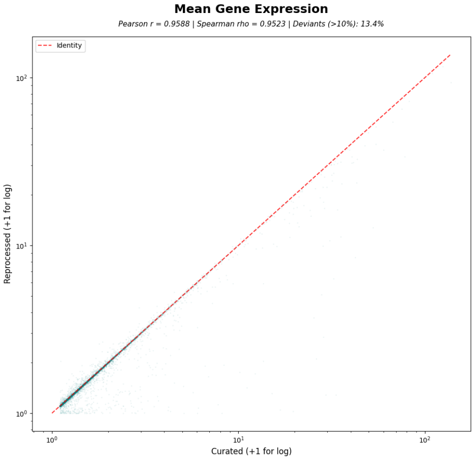


Correlation of total UMI counts per cell. Kallisto+Bustools (Reprocessed) generally identifies more UMIs per cell, especially in the high-sensitivity regime, likely due to transcript-level mapping vs gene-level quantification.

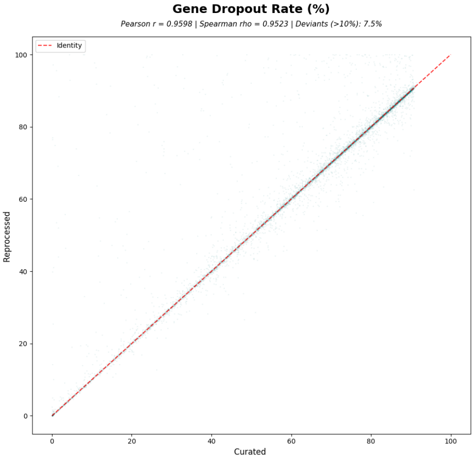


Correlation of unique genes detected per cell. The reprocessed pipeline recovers significantly more unique transcripts in a subset of cells, indicating higher detection sensitivity.

Gene-Level Expression Agreement



Average expression level for each shared gene. High correlation confirms that the biological signal is preserved across quantification methods.



Percentage of cells where a gene has zero counts. Lower values in Reprocessed indicate higher sensitivity for those genes.

Guide Recovery: Current Result

- The comparison uses a strict guide call: two positive guides, both targeting the same gene, with at least 5 counts each.
- Under this rule, the reprocessed data recovers 164,757 of 214,522 curated dual-guide cells.
- Guide-call agreement: **76.80%**.
- The missing calls are mostly cells where only one of the two curated probes is picked up strongly enough.

Interpretation: current guide recovery is about guide calling behaviour, not a failure of the expression pipeline.

Guide Signal Diagnostics

Threshold	Both curated probes	Exactly one probe	Neither probe
$\geq$ 1 count	173,043	41,332	147
$\geq$ 2 counts	170,245	44,038	239
$\geq$ 5 counts	164,752	46,288	3,482
$\geq$ 10 counts	158,854	47,254	8,414

- These counts are for curated guide-labelled shared cells.
- The threshold alone is not the main issue: even at 1 count, many cells still have only one of the two curated probes detected.
- We should test an alternative guide-calling rule that rescues a cell when one guide unambiguously points to the construct being present, and no other guide gives a conflicting signal.

What We Have Now

- A working, reproducible FASTQ-to-H5AD pipeline for perturb-seq datasets with GEX and guide modalities.
- A comparison workflow that checks cell-level expression, gene-level expression, dropout, and perturbation labels.
- First results showing very strong expression agreement with the author-processed data.
- A clear guide-calling question to tune next: how should we handle cells where one guide is strong and the second is weak or missing?

Next Steps

- **Guide recovery:** Test alternative guide-calling logic, including one-guide rescue when the signal is unambiguous.
- **Scale-out:** Run the pipeline on all available perturb-seq datasets.
- **Downstream analysis:** Run distance metrics between genes, plus the existing DEA and GSEA workflows.